PYTHON PROGRAMMING – 30 QUESTIONS

VTU Syllabus: Modules 1, 2 and 3

MODULE 1 – THE WAY OF THE PROGRAM, VARIABLES, EXPRESSIONS, ITERATION AND FUNCTIONS

1. Explain the elif, or, and while statements in Python with a suitable example of each. **[VTU PYQ – 2024 | 6 marks]**
2. List and explain the mathematical operators used in Python with examples. **[VTU PYQ – 2024 | 6 marks]**
3. Define a user-defined function. How are parameters passed to functions? Explain with an example. **[VTU PYQ – 2025 | 6 marks]**
4. What is an arithmetic expression? Predict the result of 'hello world' + 100 + 'how are you'. Explain why an error occurs, if any. **[VTU Model Question Paper | 6 marks]**
5. Write a Python program to read a person's name and year of birth and print whether the person is a senior citizen or not. **[VTU PYQ – 2024 | 6 marks]**
6. Differentiate between syntax errors, runtime errors, and semantic errors with suitable examples. **[Practice | 6 marks]**
7. Explain operator precedence and type-conversion functions in Python with suitable examples. **[Practice | 6 marks]**
8. Write a Python program to generate the Collatz 3n + 1 sequence until the value becomes 1. **[Practice | 8 marks]**
9. Explain the break and continue statements in Python with examples. **[Practice | 8 marks]**
10. Write a Python program to display multiplication tables using nested loops. **[Practice | 8 marks]**

MODULE 2 – STRINGS, TUPLES AND LISTS

11. Explain the following string methods with examples: isalpha(), isalnum(), isdecimal(), and isspace(). **[VTU PYQ – 2024 | 8 marks]**
12. Explain the list methods append(), insert(), and sort() with examples. **[VTU PYQ – 2024 | 8 marks]**
13. Explain the following list methods with examples: append(), insert(), extend(), remove(), pop(), and sort(). **[VTU PYQ – 2025 | 6 marks]**
14. Explain the use of the in and not in operators with suitable examples. **[VTU PYQ – 2025 | 6 marks]**
15. Define the tuple data type. List the differences between a tuple and a list. Name the function used to convert a list into a tuple. **[VTU PYQ – 2025 | 8 marks]**
16. Illustrate list indexing, slicing, append(), remove(), and sort() with examples. **[Previously included question | 8 marks]**
17. For a = ['hello', 'how', [1, 2, 3], [[10, 20, 30]]], determine the output of: (i) print(a[:]) (ii) print(a[-3][0]) (iii) print(a[2][:-1]) (iv) print(a[0][::-1]). **[VTU Model Question Paper | 8 marks]**
18. Using string slicing, write a Python program to reverse each word in a given string. For example, "hello how are you" should become "olleh woh era uoy". **[VTU Model Question Paper | 8 marks]**
19. Why are strings immutable in Python? Explain with an example. **[Practice | 5 marks]**
20. Explain list aliasing and cloning with a suitable Python example. **[Practice | 8 marks]**

MODULE 3 – DICTIONARIES

21. Differentiate between lists and dictionaries with examples. **[VTU PYQ – 2024 | 4 marks]**

22. Develop a Python program to print the 10 most frequently appearing words in a text file. Use a dictionary to store distinct words and their frequencies, sort by frequency in descending order, and display the first 10 entries. **[VTU PYQ – 2024 | 8 marks]**

23. Explain the dictionary methods keys(), values(), items(), and setdefault() with suitable examples. **[VTU PYQ – 2025 | 6 marks]**

24. Explain the get() and setdefault() dictionary methods with suitable examples. **[Practice | 6 marks]**

25. Discuss list and dictionary data structures with one example of each. **[VTU Model Question Paper | 6 marks]**

26. Write a Python program to read dictionary data and delete a specified key-value entry. **[VTU Model Question Paper | 7 marks]**

27. Explain how dictionaries are used as counters. Write a Python program to count the number of words in a text using a dictionary. **[VTU Python paper – 2023–24 | 10 marks]**

28. Write a Python program to count the frequency of each character in a given string using a dictionary. **[Practice | 8 marks]**

29. Explain dictionary aliasing and copying with a suitable Python example. **[Practice | 6 marks]**

30. Write a Python program to merge two dictionaries. If a key occurs in both dictionaries, add their values. **[Practice | 8 marks]**